

Problem Set I

Political Science 30100

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1 Problem Set Part I

Problem 1.1. *Let $L = ap^r(1-p)^{n-r}$. Find $\log_e L$.*

$$\begin{aligned}\log_e L &= \log_e ap^r(1-p)^{n-r} && \text{(by definition of } L\text{)} \\ &= \log_e a + \log_e p^r + \log_e (1-p)^{n-r} && (\log(ab) = \log(a) + \log(b)) \\ &= \log_e a + r \log_e p + (n-r) \log_e (1-p) && (\log(a^b) = b \log(a))\end{aligned}$$

¹With Mark Hansen